

Teacher Pack

MYP 8 · Unit 1: Real Number System, Exponents and Scientific Notation · 1.5 Square and Cube Roots

Class MYP 8A · 1 lesson · 50 minutes · mixed-ability class

Package contents (Lesson 01)

Lesson_01_Preparation	Before-lesson student sheet, 3 pages (P1–P9), 10–15 minutes
Lesson_01_Class_Worksheet	In-class student booklet, 7 pages (S, N, Examples 1–3, W, A–D, E, X)
Lesson_01_After_Lesson_Practice	Practice and self-test, 4 pages (R1, H1–H16, ST1–ST3)
Lesson_01_Teacher_Slides	Editable PowerPoint, 25 slides with presenter notes (+ PDF preview)
Mini_Whiteboard_Reference	One-page reference for both mini-whiteboard rounds
Answer_Key	Full worked solutions to every student question

Assumptions made (inputs not specified)

- **Homework time:** 25 minutes (default range 20–30). A “short on time” route is printed on the sheet.
- **School logo:** none supplied, so no logo is used. The school name appears as text in headers.
- **Calculators:** students may use a calculator to *multiply* (e.g. 7.1×7.1) but not the $\sqrt{\quad}$ or $\sqrt[3]{\quad}$ keys, except for a final “check”. Starter, mini-whiteboards and exit ticket are calculator-free.
- **Scope:** principal (non-negative) square roots only; $x^2 = 49 \Rightarrow x = \pm 7$ is *deferred* to equation work. Roots of fractions appear only as supported extension (D1).
- **Retrieval R1** (homework) assumes earlier Unit 1 lessons covered integer powers such as $(-2)^4$.
- **Estimating rule:** “the closer square/cube decides”. All numbers used in class and homework have been checked so that this rule gives the correct 1 d.p. rounding. If students ask about near-halfway cases, test the halfway value (e.g. 7.05^2).

1. Objectives and success criteria

Objective: Calculate and estimate square and cube roots.

- SC1** find the square root of a perfect square and the cube root of a perfect cube;
- SC2** find the cube root of a negative number and explain why a negative number has no real square root;
- SC3** say which two whole numbers a root lies between, and why;
- SC4** estimate a root to 1 decimal place and check by squaring or cubing.

Alignment table

Objective	Before-lesson preparation	Class modelling and practice	After-lesson questions	Exit-ticket evidence	evidence
SC1 calculate roots	P1–P3, P5, P8 (squares and cubes); P9(a)	Ex 1(a)(b); W1(a)(b); MWB1 Q1–3; A1–A5, A7, B1–B2, C1–C2	H1, H2(a)(b), H3, H5, H6, H12, ST1	X1	
SC2 negatives	P6, P7, worked example $(-2)^3$	Ex 1(c)(d); W1(c); MWB1 Q4; MWB2 Q5; A6, B3, B4	H2(c), H10, ST2	X2	
SC3 between	P4, P9(b)	N1; Ex 2–3 steps 1–3; MWB2 Q1–3; A8, B5, C3, E3	H4, H9, H11, H15	X3(a), X4	
SC4 estimate 1 d.p.	(bridge only: P9(b))	Ex 2–3; W2, W3; MWB2 Q4; B6–B8, C4, E3	H7, H8, H13, H14, H16, ST3	X3(b)	

2. Prerequisite analysis

Prerequisite	Why it matters today / where checked
Perfect squares to $15^2 = 225$	Trapping step for square roots. Preparation P1, P3, P5; Starter S1, S2.
Perfect cubes to $10^3 = 1000$	Trapping step for cube roots. P1(d)(e), P2, P5; S3, S4.
Index notation n^2 , n^3	Common error $3^3 = 9$. P2, worked example; S3.
Sign rules for multiplying negatives	$\sqrt[3]{-8} = -2$ and why $\sqrt{-9}$ is not real. P6, P7; S5.
Squaring/cubing decimals (calculator)	The Test step. Calculator allowed for multiplying.
Area of a square, volume of a cube	Context and meaning (side from area, edge from volume). P8, P9.

Key vocabulary (same wording on worksheet p. 1 and slide 5)

square root $\sqrt{n}$: the non-negative number that multiplied by itself gives n . **cube root** $\sqrt[3]{n}$: the number that cubed gives n . **radical sign**. **perfect square** / **perfect cube**. **inverse operation**. **estimate** ($\approx$).

Resources

Mini-whiteboards, pens and erasers (one per student); printed Class Worksheet (7 pages, stapled; page 7 exit ticket detachable); spare copies of Preparation page 1 (Remember box) for students who did not complete preparation; calculators; Teacher Slides; After-Lesson Practice copies; board space divided as in the board plan.

3. Minute-by-minute lesson plan (50 minutes)

Time	Stage	Teacher action	Student action	Resource	Evidence / decision
0:00–0:01	Entry	Greet; slide 1 objective; hand out worksheet; give Remember box to anyone without preparation	Write name/date; read objective	Slide 1; WS p.1	Who lacks preparation?
0:01–0:04	Starter	Circulate; note errors on 3^3 and $(-4)^2$	S1–S6 alone, no calculator	Slide 2; WS p.1	Scan for $3^3 = 9$, $(-4)^2 = -16$
0:04–0:05	Starter feedback	Reveal; hands up for 5–6 correct	Self-mark, correct	Slide 3	If many miss S3/S5, 30 s on board
0:05–0:08	Prerequisite check / prep link	Discuss P9; fingers for side of 36; reveal card; set N1	Answer; complete N1	Slide 4; WS p.1 N1	Can they say “between 4 and 5”?
0:08–0:10	I DO: concept	Define $\sqrt{\quad}$ and $\sqrt[3]{\quad}$ with square/cube pictures; write vocabulary	Listen, answer “side / edge” questions	Slide 5; WS p.1	Correct use of words
0:10–0:12	I DO Ex 1	Model (a)–(d) with checks	Annotate WS p.2	Slide 6	Answer “why not $125 \div 3$?”
0:12–0:15	I DO Ex 2	Model $\sqrt{50}$: trap, roots, closer, test, decide	Annotate; answer “closer?”	Slides 7–8; WS p.2	Can they say which is closer?
0:15–0:17	I DO Ex 3 + method	Model $\sqrt[3]{30}$; summarise five steps	Annotate; read Support Box	Slides 9–10; WS p.2–3	Use cubes, not squares
0:17–0:21	MWB Round 1	Routine; 5 questions; scan; decide	Boards: Q1–Q5	Slides 11–12	$\geq 80\%$ on Q3–Q5 → WE DO; else 2–3 min reteach + re-check
0:21–0:23	WE DO W1	Full support; prompt each blank	Fill blanks	Slides 13–14; WS p.3	(b) 4 not 8
0:23–0:25:30	WE DO W2	Partial support; class supplies each step	Complete template	Slides 15–16; WS p.3	Test step used?
0:25:30–0:28	WE DO W3	Minimal support; pairs compare	Five steps alone, then pair	Slides 17–18; WS p.3	Cubes chosen correctly?
0:28–0:31	MWB Round 2	Routine; scan; decide groups for YOU DO	Boards: Q1–Q5	Slides 19–20	$\geq 80\%$ → start A8/B; else reteach $\sqrt{70}$; flagged students to front table
0:31–0:41	YOU DO	3 min with flagged group; then circulate; 2-min warning at 0:39	Must A1–A8, B1–B4; Should B5–B8; Could C; Ext D	Slide 21; WS p.4–5	Check B4 wording, B6–B8 tests
0:41–0:44	Check for understanding	Pairs 90 s on E1–E3; cold-call; reveal	Find first error, explain, correct	Slides 22–23; WS p.6	Can they name the misconception?
0:44–0:48	Exit ticket	Silent; no help; collect	X1–X4 alone	Slide 24; WS p.7	Sort into 3 piles (section 11)
0:48–0:50	Recap and homework	Five steps recalled; hand out practice	Say steps to partner	Slide 25	—

Stage totals: Starter 5 + Prerequisite 3 + I DO 9 + MWB1 4 + WE DO 7 + MWB2 3 + YOU DO 10 + Check 3 + Exit 4 + Recap 2 = **50** minutes. Transitions are built into each stage (boards out/in, page turns).

4. Preparation follow-up

- **Before the lesson:** glance at self-check ticks and “My question” if sheets were handed in; otherwise ask for a show of hands at 0:05.
- **Most common preparation errors** (two-minute response on slide 4): $3^3 = 9$ (write $3 \times 3 \times 3$ in full); $(-3)^3 = 27$ (sign rule: two negatives make a positive, a third makes it negative again); P9(b) answered “20” (halving). Response: “Area 40 means side $\times$ side = 40. Try 6×6 and 7×7 .”
- **Students who did not complete preparation:** give them preparation page 1 (Remember box and list of squares/cubes) during the starter. Slide 4 is designed as their entry point; N1 on worksheet p. 1 repeats the key idea.

5. Slide / worksheet cross-reference

Slides	Content	Worksheet
1	Title, objective, success criteria	p.1
2–3	Starter and answers	p.1, S1–S6
4	Link to preparation P9	p.1, N1
5	Vocabulary and concept visual	p.1 vocabulary
6–9	I DO Examples 1, 2 (two slides), 3	p.2
10	Five-step method	p.3 Support Box
11–12	Mini-whiteboard Round 1 and answers	—
13–18	WE DO W1–W3 with separate solution slides	p.3
19–20	Mini-whiteboard Round 2 and answers	—
21	YOU DO instructions	p.4–5, A–D
22–23	Spot the Mistake and corrections	p.6, E1–E3
24	Exit ticket	p.7, X1–X4
25	Recap and homework	After-Lesson Practice

6. Board plan

Square and Cube Roots Obj: calculate and estimate $\sqrt{\quad}$ and $\sqrt[3]{\quad}$ Vocabulary square root $\sqrt{\quad}$ cube root $\sqrt[3]{\quad}$ perfect square / cube inverse operation estimate $\approx$ Remember $\sqrt{n}$ is never negative $\sqrt{\text{neg}}$: not real $\sqrt[3]{\text{neg}}$: negative Squares: 1 4 9 16 25 36 49 64 81 100 121 144 Cubes: 1 8 27 64 125 216 343 512 729 1000	Method 1. Trap 2. Roots 3. Closer? 4. Test 5. Decide ($\approx$) Ex 1 $\sqrt{81} = 9$ ($9^2 = 81$) $\sqrt[3]{125} = 5$ ($5^3 = 125$) $\sqrt[3]{-8} = -2$ ($(-2)^3 = -8$) $\sqrt{-9}$ not real Ex 2 $49 < 50 < 64$ $7 < \sqrt{50} < 8$, nearer 7 $7.1^2 = 50.41$ so $\sqrt{50} \approx 7.1$ Ex 3 $27 < 30 < 64$, $3 < \sqrt[3]{30} < 4$ $3.1^3 = 29.791$ so $\sqrt[3]{30} \approx 3.1$	Common mistake $\sqrt{36} \neq 36 \div 2$ $\sqrt[3]{27} \neq 27 \div 3$ $\sqrt{9+16} \neq 3+4$ Check by squaring/cubing! Your Turn Must: A1–A8, B1–B4 Should: B5–B8 Could: C Ext: D Sentence stem “$\sqrt{\quad}$ is between $\quad$ and $\quad$ because $\quad < \quad < \quad$” Homework: After-Lesson Practice (25 min)
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7. Teaching script

Teaching point 1: What a square root is (slides 4–5, 0:05–0:10)

SAY	“Finding the side of a square from its area undoes squaring. We call it the square root. $\sqrt{n}$ is the number, not negative, that multiplied by itself gives n .”
WRITE	$\sqrt{49} = 7$ because $7^2 = 49$. Vocabulary list.
ASK	“A square has area 36. How long is its side? How do you know?”
EXPECT	“6, because $6 \times 6 = 36$.”
WATCH FOR	“18” (halving); reading $\sqrt{\quad}$ as “divide”.
IF STUDENTS STRUGGLE	Draw a 6×6 grid, count squares. Ask “what <i>side</i> makes this area?” not “what is half?”

Teaching point 2: Cube roots, including negatives (slides 5–6, 0:08–0:12)

SAY	“The small 3 makes a cube root. It undoes cubing: the edge of a cube from its volume. Negative numbers <i>do</i> have cube roots, because a negative cubed is negative. But no real number squares to a negative, so $\sqrt{-9}$ is not a real number.”
WRITE	$\sqrt[3]{125} = 5$ ($5 \times 5 \times 5 = 125$); $\sqrt[3]{-8} = -2$ ($(-2)^3 = -8$); $\sqrt{-9}$: not real.
ASK	“Could $\sqrt{-9}$ be -3 ? Check: what is $(-3)^2$?”
EXPECT	$(-3)^2 = 9$, not -9 . So no.”
WATCH FOR	$\sqrt[3]{125} = 41.7$ (dividing by 3); $\sqrt[3]{-8} = \text{“no answer”}$; $\sqrt{-9} = -3$.
IF STUDENTS STRUGGLE	Use the checking habit: “Cube your answer. Do you get the number back?” Show $(-2) \times (-2) = 4$, $4 \times (-2) = -8$ step by step.

Teaching point 3: Trapping a root between whole numbers (slide 7, 0:12–0:13:30)

SAY	“50 is not a perfect square, so $\sqrt{50}$ is not a whole number. We trap 50 between the perfect squares either side, then take the square root of every part.”
WRITE	$49 < 50 < 64 \Rightarrow 7 < \sqrt{50} < 8$
ASK	“Is $\sqrt{50}$ nearer to 7 or to 8? How can you tell without a calculator?”
EXPECT	“Nearer 7, because 50 is only 1 more than 49 but 14 less than 64.”
WATCH FOR	Writing $49 < \sqrt{50} < 64$; choosing non-neighbouring squares (e.g. 36 and 64).
IF STUDENTS STRUGGLE	Write the list of squares on the board; point to 50’s position in the list; “which two squares is 50 sitting between?”

Teaching point 4: Estimating to 1 d.p. and checking (slides 8–10, 0:13:30–0:17)

SAY	“Now we test decimals. Square 7.1: 50.41, just above 50. $7.0^2 = 49$, below. So $\sqrt{50}$ is between 7.0 and 7.1. 50 is closer to 50.41, so $\sqrt{50} \approx 7.1$. For cube roots we do the same with cubes.”
WRITE	$7.0^2 = 49$, $7.1^2 = 50.41 \Rightarrow \sqrt{50} \approx 7.1$; $3.1^3 = 29.791$, $3.2^3 = 32.768 \Rightarrow \sqrt[3]{30} \approx 3.1$. Five-step method.
ASK	“Which squared number is closer to 50: 49 or 50.41?” “Which perfect cubes trap 30?”
EXPECT	“50.41, only 0.41 away.” “27 and 64.”
WATCH FOR	Stopping at “between 7 and 8”; using squares for a cube root; “halfway” reasoning ($\sqrt{40} \approx 6.5$).
IF STUDENTS STRUGGLE	Give the first test value (“try 7.1”), and a two-column table “decimal — squared”. Allow calculator for multiplying.

8. Mini-whiteboard rounds

Routine (both rounds): “Boards down $\rightarrow$ work independently $\rightarrow$ show in 3, 2, 1 $\rightarrow$ do not erase $\rightarrow$ scan all boards.” Ask one question at a time. **Decision threshold:** about 80% of boards correct on the harder questions (Q3–Q5).

Round 1 (slides 11–12, 0:17–0:21): calculate

Q	Question	Answer	Misconception / check	If many miss it
1 (very easy)	$\sqrt{9}$	3	4.5 (halving)	“Which number times itself is 9?”
2 (easy)	$\sqrt[3]{8}$	2	$\frac{8}{3}$ or 4	Draw a $2 \times 2 \times 2$ cube; count 8.
3 (similar)	$\sqrt{64}$	8	4 (confuses with $\sqrt[3]{64}$)	Point to the (missing) small 3: “square or cube root?”
4 (variation)	$\sqrt[3]{-125}$	–5	“no answer” or 5	Check $(-5)^3$ step by step on the board.
5 (misconception)	True/false: $\sqrt[3]{27} = 9$	False, 3	dividing by 3	“Cube 9: $9 \times 9 \times 9 = 729$. Not 27.”

Below threshold: 2–3 minute reteach with the $3 \times 3 \times 3$ cube picture: “ $\sqrt[3]{27}$ asks for the edge, not the volume divided by 3.” **Re-check:** $\sqrt{36}$ (6) and $\sqrt[3]{1000}$ (10). Take the 2–3 minutes from the start of YOU DO.

Round 2 (slides 19–20, 0:28–0:31): between and estimate

Q	Question	Answer	Misconception / check	If many miss it
1 (very easy)	$\sqrt{10}$ between?	3 and 4	9 and 16 (forgot to root)	“Now take the root of each end.”
2 (easy)	$\sqrt{45}$ between?	6 and 7	5 and 7 (non-neighbours)	Use the list: 36, 45 , 49.
3 (similar)	$\sqrt[3]{20}$ between?	2 and 3	4 and 5 (used squares)	“Is 16 a cube? List the cubes.”
4 (variation)	$\sqrt{30}$ to 1 d.p.	5.5	5.4 or “between 5 and 6”	$5.4^2 = 29.16$, $5.5^2 = 30.25$; which is closer?
5 (misconception)	True/false: $\sqrt{-4} = -2$	False; not real	$(-2)^2 = -4$	$(-2) \times (-2) = +4$; contrast with $\sqrt[3]{-8} = -2$.

Below threshold: 2–3 minute reteach with $\sqrt{70}$: trap $64 < 70 < 81$, roots 8 and 9, closer to 64, test $8.3^2 = 68.89$, $8.4^2 = 70.56$, so $\sqrt{70} \approx 8.4$. **Re-check:** “ $\sqrt{12}$ is between?” (3 and 4). Students who still miss it start YOU DO at A1 with you at the front table.

9. WE DO with fading support (slides 13–18, 0:21–0:28)**W1: full support (fill in the blanks)**

Prompts: “What times itself is 144?” “Careful, cube root: what cubed is 64?” “Positive or negative answer for $\sqrt[3]{-27}$? Why?”

Expected: 12; 4 (not 8); –3 “because a negative cubed is negative”.

Solution: (a) $\sqrt{144} = 12$ since $12^2 = 144$. (b) $\sqrt[3]{64} = 4$ since $4 \times 4 \times 4 = 64$. (c) $\sqrt[3]{-27} = -3$ since $(-3)^3 = -27$.

W2: partial support (template)

Prompts: “Trap?” “Roots?” “Closer to?” “Square 4.4 and 4.5.” “Which is closer to 20?”

Expected: $16 < 20 < 25$; $4 < \sqrt{20} < 5$; “slightly nearer 16” (4 away vs 5 away: a close call, so $\sqrt{20}$ is near the middle and the test decides); 19.36 and 20.25; 20.25.

Solution: $4.4^2 = 19.36 < 20 < 20.25 = 4.5^2$. $20 - 19.36 = 0.64$, $20.25 - 20 = 0.25$, so $\sqrt{20} \approx 4.5$.

W3: minimal support (five steps, then compare with a partner)**Prompt:** “Use the five steps. Is 100 a perfect *cube*?”**Expected:** $64 < 100 < 125$; $4 < \sqrt[3]{100} < 5$; 36 from 64, 25 from 125, so try the middle; $4.6^3 = 97.336$, $4.7^3 = 103.823$.**Solution:** $100 - 97.336 = 2.664$ and $103.823 - 100 = 3.823$, so $\sqrt[3]{100} \approx 4.6$. (Watch for $\sqrt{100} = 10$ confusion.)**10. Differentiation (mixed-ability class)**

Needs more support	Remember box / Support Box list of squares and cubes; front-table group for the first 3 minutes of YOU DO; start at A1; sentence stem; calculator for multiplying; W2 template reused for B6–B8.
Core	Must + Should (A1–B8); explanation B4 in full sentences.
Ready for more	Start at A8 if all correct in Round 2; Section C then D; ask for a written justification of C5 and D2 using the words “because” and “check”.
EAL	Vocabulary box; model the sentence stem aloud; accept a diagram plus key words for B4.

11. Misconceptions

Misconception	Example	Diagnosis	Teacher response
Square root means halve	$\sqrt{36} = 18$	Confuses “root” with the 2 in n^2	“Check: $18^2 = 324$.” Area/side picture. (E1, MWB1 Q1)
Cube root means divide by 3	$\sqrt[3]{27} = 9$	Same confusion with the index 3	Cube the answer: $9^3 = 729$. (MWB1 Q5)
Negative roots	$\sqrt{-4} = -2$; $\sqrt[3]{-8}$ “impossible”	Sign rules not applied; treats $\sqrt{\quad}$ and $\sqrt[3]{\quad}$ alike	$(-2)^2 = +4$ but $(-2)^3 = -8$. (Ex 1, MWB1 Q4, MWB2 Q5)
Root of a sum	$\sqrt{9 + 16} = 3 + 4$	Over-generalises distributing	Evaluate inside first: $\sqrt{25} = 5 \neq 7$. (E2)
Halfway reasoning	$\sqrt{40} \approx 6.5$	Assumes roots are spaced evenly	Measure distances: 4 vs 9; test 6.3^2 . (E3, H14)
Squares for cube roots	$\sqrt[3]{20}$ between 4 and 5	Uses the familiar list	“Is 16 a cube?” List cubes first. (MWB2 Q3, B8)

12. Exit ticket interpretation and next-lesson decisions

The essential objective is judged on **X1–X3** (calculate and estimate). X4 is slight transfer (context + cube roots).

Most secure	students (X1–X3 correct, $\geq 80\%$)	Next lesson: 3-question retrieval starter (one calculate, one estimate, one negative cube root), then move on in Unit 1. Students who missed X4 get H13/H16-style context questions as starter.
About half struggling		Re-group: struggling half gets a 10-minute guided session on the five steps (reuse W2 template with $\sqrt{30}$, $\sqrt[3]{50}$), secure half does Section C/D and homework Hard questions. Re-check with two mini-whiteboard questions before continuing.
Most struggling		Revisit prerequisites first: perfect squares/cubes fluency (5 minutes), then re-teach calculating roots with area/volume pictures before estimating. Re-run Round 2 (different numbers: $\sqrt{17}$, $\sqrt{60}$, $\sqrt[3]{12}$) and delay new content by one lesson.

Diagnosing by question: X1 wrong $\rightarrow$ perfect squares/cubes not known (prerequisite). X2 wrong $\rightarrow$ sign rules. X3(a) right but (b) wrong $\rightarrow$ Test step. X4 wrong but X3 right $\rightarrow$ using squares for cube roots or not linking volume to cube root.